

Table 1: Complete performance comparison across algorithms and step budgets.

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
SafePO	1c3s5z	7.64 $\pm$ 0.16	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	1c3s5z	7.64 $\pm$ 0.08	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	1c3s5z	8.08 $\pm$ 0.21	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	1c3s5z	9.09 $\pm$ 0.76	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	1c3s5z	9.52 $\pm$ 0.61	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	1c3s5z	11.05 $\pm$ 0.21	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	1c3s5z	17.54 $\pm$ 0.94	7.03 $\pm$ 1.70	0.36 $\pm$ 0.27	500k	0
MAPPO-Lag	1c3s5z	19.82 $\pm$ 0.15	2.22 $\pm$ 0.64	0.96 $\pm$ 0.04	1M	0
MAPPO-Lag	1c3s5z	19.96 $\pm$ 0.01	1.04 $\pm$ 0.04	1.00 $\pm$ 0.00	2M	0
MAPPO-Lag	1c3s5z	20.00 $\pm$ 0.00	0.44 $\pm$ 0.11	1.00 $\pm$ 0.00	5M	0
Safe Dreamers	1c3s5z	19.29 $\pm$ 0.14	4.23 $\pm$ 0.27	0.82 $\pm$ 0.02	100k	0
SafePO	1c3s5z	7.79 $\pm$ 0.05	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4
SafePO	1c3s5z	7.76 $\pm$ 0.06	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
SafePO	1c3s5z	8.96 $\pm$ 0.70	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
SafePO	1c3s5z	10.02 $\pm$ 0.18	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	4
SafePO	1c3s5z	12.36 $\pm$ 0.37	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	4
MAPPO-Lag	1c3s5z	11.04 $\pm$ 0.07	9.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
MAPPO-Lag	1c3s5z	17.24 $\pm$ 1.16	7.19 $\pm$ 1.71	0.32 $\pm$ 0.30	500k	4
MAPPO-Lag	1c3s5z	19.82 $\pm$ 0.16	2.08 $\pm$ 0.66	0.96 $\pm$ 0.04	1M	4
MAPPO-Lag	1c3s5z	19.95 $\pm$ 0.01	0.98 $\pm$ 0.18	1.00 $\pm$ 0.00	2M	4
MAPPO-Lag	1c3s5z	20.00 $\pm$ 0.00	0.37 $\pm$ 0.14	1.00 $\pm$ 0.00	5M	4
Safe Dreamers	1c3s5z	19.15 $\pm$ 0.53	4.83 $\pm$ 0.94	0.79 $\pm$ 0.12	100k	4
SafePO	2m_vs_1z	2.71 $\pm$ 0.18	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	2m_vs_1z	2.73 $\pm$ 0.16	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	2m_vs_1z	4.58 $\pm$ 0.11	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	2m_vs_1z	5.99 $\pm$ 0.60	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	2m_vs_1z	6.96 $\pm$ 0.76	2.00 $\pm$ 0.01	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	2m_vs_1z	4.71 $\pm$ 0.06	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	2m_vs_1z	19.97 $\pm$ 0.02	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	500k	0
MAPPO-Lag	2m_vs_1z	19.95 $\pm$ 0.01	0.01 $\pm$ 0.00	1.00 $\pm$ 0.00	1M	0
MAPPO-Lag	2m_vs_1z	20.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	2M	0
MAPPO-Lag	2m_vs_1z	20.00 $\pm$ 0.01	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	5M	0
Safe Dreamers	2m_vs_1z	19.30 $\pm$ 0.49	0.14 $\pm$ 0.08	0.95 $\pm$ 0.04	100k	0
SafePO	2m_vs_1z	3.13 $\pm$ 0.01	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
SafePO	2m_vs_1z	3.09 $\pm$ 0.04	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
SafePO	2m_vs_1z	5.04 $\pm$ 0.07	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	1
SafePO	2m_vs_1z	8.52 $\pm$ 3.15	1.76 $\pm$ 0.35	0.15 $\pm$ 0.21	1M	1
SafePO	2m_vs_1z	19.89 $\pm$ 0.06	0.02 $\pm$ 0.01	0.99 $\pm$ 0.01	2M	1
MAPPO-Lag	2m_vs_1z	4.68 $\pm$ 0.09	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
MAPPO-Lag	2m_vs_1z	19.99 $\pm$ 0.01	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	500k	1
MAPPO-Lag	2m_vs_1z	19.96 $\pm$ 0.00	0.01 $\pm$ 0.00	1.00 $\pm$ 0.00	1M	1
MAPPO-Lag	2m_vs_1z	20.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	2M	1
MAPPO-Lag	2m_vs_1z	20.00 $\pm$ 0.01	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	5M	1
Safe Dreamers	2m_vs_1z	19.34 $\pm$ 0.38	0.14 $\pm$ 0.09	0.95 $\pm$ 0.03	100k	1
SafePO	2s3z	3.69 $\pm$ 0.02	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	2s3z	3.67 $\pm$ 0.03	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	2s3z	3.19 $\pm$ 0.16	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	2s3z	0.99 $\pm$ 0.15	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	2s3z	0.27 $\pm$ 0.05	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	2s3z	7.14 $\pm$ 0.19	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	2s3z	14.54 $\pm$ 1.35	4.42 $\pm$ 0.60	0.20 $\pm$ 0.21	500k	0

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Table 1 – continued

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
MAPPO-Lag	2s3z	18.15 $\pm$ 1.96	2.88 $\pm$ 1.86	0.59 $\pm$ 0.51	1M	0
MAPPO-Lag	2s3z	18.75 $\pm$ 1.91	2.09 $\pm$ 2.54	0.66 $\pm$ 0.57	2M	0
MAPPO-Lag	2s3z	19.95 $\pm$ 0.05	0.28 $\pm$ 0.15	0.99 $\pm$ 0.01	5M	0
Safe Dreamers	2s3z	18.44 $\pm$ 0.25	2.28 $\pm$ 0.38	0.77 $\pm$ 0.04	100k	0
SafePO	2s3z	4.86 $\pm$ 0.04	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4
SafePO	2s3z	4.85 $\pm$ 0.03	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
SafePO	2s3z	7.61 $\pm$ 0.09	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
SafePO	2s3z	8.85 $\pm$ 0.16	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	4
SafePO	2s3z	11.86 $\pm$ 0.77	4.95 $\pm$ 0.09	0.02 $\pm$ 0.03	2M	4
MAPPO-Lag	2s3z	7.14 $\pm$ 0.31	5.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
MAPPO-Lag	2s3z	13.66 $\pm$ 1.30	4.68 $\pm$ 0.48	0.11 $\pm$ 0.17	500k	4
MAPPO-Lag	2s3z	18.57 $\pm$ 1.52	2.26 $\pm$ 1.10	0.74 $\pm$ 0.30	1M	4
MAPPO-Lag	2s3z	19.79 $\pm$ 0.10	0.59 $\pm$ 0.12	0.97 $\pm$ 0.02	2M	4
MAPPO-Lag	2s3z	19.96 $\pm$ 0.02	0.18 $\pm$ 0.06	0.99 $\pm$ 0.01	5M	4
Safe Dreamers	2s3z	17.37 $\pm$ 1.38	2.98 $\pm$ 0.64	0.63 $\pm$ 0.15	100k	4
SafePO	2s_vs_1sc	2.44 $\pm$ 0.10	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	2s_vs_1sc	2.59 $\pm$ 0.16	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	2s_vs_1sc	2.31 $\pm$ 0.12	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	2s_vs_1sc	2.14 $\pm$ 1.15	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	2s_vs_1sc	0.10 $\pm$ 0.05	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	2s_vs_1sc	6.05 $\pm$ 1.62	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	2s_vs_1sc	19.69 $\pm$ 0.47	0.62 $\pm$ 0.25	0.93 $\pm$ 0.07	500k	0
MAPPO-Lag	2s_vs_1sc	20.17 $\pm$ 0.01	0.21 $\pm$ 0.01	1.00 $\pm$ 0.00	1M	0
MAPPO-Lag	2s_vs_1sc	20.23 $\pm$ 0.00	0.17 $\pm$ 0.06	1.00 $\pm$ 0.00	2M	0
MAPPO-Lag	2s_vs_1sc	20.23 $\pm$ 0.01	0.11 $\pm$ 0.07	1.00 $\pm$ 0.00	5M	0
Safe Dreamers	2s_vs_1sc	19.95 $\pm$ 0.21	0.23 $\pm$ 0.11	0.96 $\pm$ 0.03	100k	0
SafePO	2s_vs_1sc	4.21 $\pm$ 0.21	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
SafePO	2s_vs_1sc	4.08 $\pm$ 0.21	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
SafePO	2s_vs_1sc	11.93 $\pm$ 0.49	1.89 $\pm$ 0.04	0.10 $\pm$ 0.04	500k	1
SafePO	2s_vs_1sc	16.49 $\pm$ 0.81	1.34 $\pm$ 0.12	0.55 $\pm$ 0.09	1M	1
SafePO	2s_vs_1sc	19.34 $\pm$ 0.13	0.78 $\pm$ 0.04	0.89 $\pm$ 0.01	2M	1
MAPPO-Lag	2s_vs_1sc	6.06 $\pm$ 1.61	2.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
MAPPO-Lag	2s_vs_1sc	18.79 $\pm$ 1.35	0.94 $\pm$ 0.23	0.83 $\pm$ 0.16	500k	1
MAPPO-Lag	2s_vs_1sc	20.17 $\pm$ 0.01	0.18 $\pm$ 0.04	1.00 $\pm$ 0.00	1M	1
MAPPO-Lag	2s_vs_1sc	20.23 $\pm$ 0.00	0.13 $\pm$ 0.02	1.00 $\pm$ 0.00	2M	1
MAPPO-Lag	2s_vs_1sc	20.23 $\pm$ 0.00	0.10 $\pm$ 0.03	1.00 $\pm$ 0.00	5M	1
Safe Dreamers	2s_vs_1sc	19.57 $\pm$ 0.49	0.31 $\pm$ 0.10	0.92 $\pm$ 0.06	100k	1
SafePO	3m	2.06 $\pm$ 0.10	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	3m	2.04 $\pm$ 0.13	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	3m	2.88 $\pm$ 0.12	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	3m	1.04 $\pm$ 0.16	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	3m	0.58 $\pm$ 0.15	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3m	4.37 $\pm$ 0.31	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	3m	18.76 $\pm$ 0.90	0.92 $\pm$ 0.58	0.90 $\pm$ 0.07	500k	0
MAPPO-Lag	3m	19.90 $\pm$ 0.05	0.16 $\pm$ 0.13	0.99 $\pm$ 0.00	1M	0
MAPPO-Lag	3m	19.96 $\pm$ 0.04	0.08 $\pm$ 0.10	1.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3m	19.99 $\pm$ 0.01	0.02 $\pm$ 0.01	1.00 $\pm$ 0.00	5M	0
Safe Dreamers	3m	19.57 $\pm$ 0.25	0.35 $\pm$ 0.14	0.97 $\pm$ 0.02	100k	0
SafePO	3m	2.55 $\pm$ 0.04	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
SafePO	3m	2.54 $\pm$ 0.06	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
SafePO	3m	7.14 $\pm$ 0.16	2.86 $\pm$ 0.02	0.08 $\pm$ 0.01	500k	1
SafePO	3m	15.05 $\pm$ 1.29	1.83 $\pm$ 0.25	0.62 $\pm$ 0.10	1M	1
SafePO	3m	19.08 $\pm$ 0.04	1.03 $\pm$ 0.16	0.93 $\pm$ 0.00	2M	1

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Table 1 – continued

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
MAPPO-Lag	3m	4.43 $\pm$ 0.32	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
MAPPO-Lag	3m	19.83 $\pm$ 0.16	0.19 $\pm$ 0.14	0.99 $\pm$ 0.01	500k	1
MAPPO-Lag	3m	19.93 $\pm$ 0.02	0.05 $\pm$ 0.01	1.00 $\pm$ 0.00	1M	1
MAPPO-Lag	3m	19.99 $\pm$ 0.01	0.02 $\pm$ 0.00	1.00 $\pm$ 0.00	2M	1
MAPPO-Lag	3m	20.00 $\pm$ 0.00	0.02 $\pm$ 0.00	1.00 $\pm$ 0.00	5M	1
Safe Dreamers	3m	19.46 $\pm$ 0.42	0.60 $\pm$ 0.17	0.96 $\pm$ 0.03	100k	1
SafePO	3s5z_vs_3s6z	3.47 $\pm$ 0.02	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	3s5z_vs_3s6z	3.45 $\pm$ 0.03	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	3s5z_vs_3s6z	3.24 $\pm$ 0.02	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	3s5z_vs_3s6z	2.88 $\pm$ 0.24	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	3s5z_vs_3s6z	2.53 $\pm$ 0.17	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3s5z_vs_3s6z	5.08 $\pm$ 0.01	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	3s5z_vs_3s6z	9.27 $\pm$ 0.59	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
MAPPO-Lag	3s5z_vs_3s6z	10.46 $\pm$ 0.56	8.07 $\pm$ 0.01	0.00 $\pm$ 0.00	1M	0
MAPPO-Lag	3s5z_vs_3s6z	11.75 $\pm$ 1.03	8.09 $\pm$ 0.01	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3s5z_vs_3s6z	13.42 $\pm$ 0.66	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s5z_vs_3s6z	9.60 $\pm$ 0.84	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	3s5z_vs_3s6z	3.86 $\pm$ 0.04	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4
SafePO	3s5z_vs_3s6z	3.86 $\pm$ 0.02	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
SafePO	3s5z_vs_3s6z	5.14 $\pm$ 0.03	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
SafePO	3s5z_vs_3s6z	5.62 $\pm$ 0.04	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	4
SafePO	3s5z_vs_3s6z	6.08 $\pm$ 1.09	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	4
MAPPO-Lag	3s5z_vs_3s6z	5.08 $\pm$ 0.11	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
MAPPO-Lag	3s5z_vs_3s6z	9.53 $\pm$ 0.26	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
MAPPO-Lag	3s5z_vs_3s6z	10.57 $\pm$ 0.05	8.08 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	4
MAPPO-Lag	3s5z_vs_3s6z	11.75 $\pm$ 0.45	8.07 $\pm$ 0.02	0.00 $\pm$ 0.00	2M	4
MAPPO-Lag	3s5z_vs_3s6z	13.10 $\pm$ 1.06	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4
Safe Dreamers	3s5z_vs_3s6z	13.28 $\pm$ 1.10	7.54 $\pm$ 0.40	0.11 $\pm$ 0.06	100k	4
SafePO	3s_vs_3z	6.04 $\pm$ 0.04	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	3s_vs_3z	5.99 $\pm$ 0.07	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	3s_vs_3z	8.12 $\pm$ 0.18	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	3s_vs_3z	5.45 $\pm$ 0.47	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	3s_vs_3z	2.94 $\pm$ 1.89	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3s_vs_3z	8.07 $\pm$ 0.05	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	3s_vs_3z	10.44 $\pm$ 0.18	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
MAPPO-Lag	3s_vs_3z	12.03 $\pm$ 0.39	3.02 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
MAPPO-Lag	3s_vs_3z	19.81 $\pm$ 0.08	0.12 $\pm$ 0.06	0.98 $\pm$ 0.01	2M	0
MAPPO-Lag	3s_vs_3z	20.01 $\pm$ 0.02	0.01 $\pm$ 0.01	1.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s_vs_3z	19.83 $\pm$ 0.20	0.95 $\pm$ 0.18	0.96 $\pm$ 0.02	100k	0
SafePO	3s_vs_3z	6.28 $\pm$ 0.10	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
SafePO	3s_vs_3z	6.25 $\pm$ 0.04	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
SafePO	3s_vs_3z	8.08 $\pm$ 0.03	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	1
SafePO	3s_vs_3z	9.76 $\pm$ 0.11	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	1
SafePO	3s_vs_3z	11.68 $\pm$ 0.58	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	1
MAPPO-Lag	3s_vs_3z	8.06 $\pm$ 0.06	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
MAPPO-Lag	3s_vs_3z	10.66 $\pm$ 0.31	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	1
MAPPO-Lag	3s_vs_3z	12.01 $\pm$ 0.32	3.01 $\pm$ 0.01	0.00 $\pm$ 0.00	1M	1
MAPPO-Lag	3s_vs_3z	19.64 $\pm$ 0.13	0.21 $\pm$ 0.04	0.96 $\pm$ 0.02	2M	1
MAPPO-Lag	3s_vs_3z	20.02 $\pm$ 0.04	0.01 $\pm$ 0.01	1.00 $\pm$ 0.00	5M	1
Safe Dreamers	3s_vs_3z	19.93 $\pm$ 0.20	0.79 $\pm$ 0.48	0.96 $\pm$ 0.04	100k	1
SafePO	3s_vs_4z	4.39 $\pm$ 0.03	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	3s_vs_4z	4.38 $\pm$ 0.03	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	3s_vs_4z	5.71 $\pm$ 0.12	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0

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Table 1 – continued

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
SafePO	3s_vs_4z	3.86 $\pm$ 0.42	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	3s_vs_4z	2.46 $\pm$ 0.08	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3s_vs_4z	5.66 $\pm$ 0.17	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	3s_vs_4z	6.83 $\pm$ 0.12	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
MAPPO-Lag	3s_vs_4z	7.49 $\pm$ 0.04	3.01 $\pm$ 0.01	0.00 $\pm$ 0.00	1M	0
MAPPO-Lag	3s_vs_4z	9.36 $\pm$ 0.39	3.03 $\pm$ 0.01	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3s_vs_4z	13.07 $\pm$ 0.86	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s_vs_4z	18.19 $\pm$ 1.57	2.64 $\pm$ 0.34	0.23 $\pm$ 0.16	100k	0
SafePO	3s_vs_4z	4.55 $\pm$ 0.01	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
SafePO	3s_vs_4z	4.56 $\pm$ 0.00	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
SafePO	3s_vs_4z	5.78 $\pm$ 0.01	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	1
SafePO	3s_vs_4z	7.05 $\pm$ 0.18	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	1
SafePO	3s_vs_4z	8.38 $\pm$ 0.33	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	1
MAPPO-Lag	3s_vs_4z	5.63 $\pm$ 0.06	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
MAPPO-Lag	3s_vs_4z	6.87 $\pm$ 0.10	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	1
MAPPO-Lag	3s_vs_4z	7.73 $\pm$ 0.17	3.02 $\pm$ 0.01	0.00 $\pm$ 0.00	1M	1
MAPPO-Lag	3s_vs_4z	9.35 $\pm$ 0.12	3.02 $\pm$ 0.01	0.00 $\pm$ 0.00	2M	1
MAPPO-Lag	3s_vs_4z	12.52 $\pm$ 1.70	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
Safe Dreamers	3s_vs_4z	16.96 $\pm$ 0.74	2.47 $\pm$ 0.35	0.31 $\pm$ 0.11	100k	1
SafePO	3s_vs_5z	3.44 $\pm$ 0.03	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	3s_vs_5z	3.41 $\pm$ 0.02	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	3s_vs_5z	4.58 $\pm$ 0.24	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	3s_vs_5z	4.43 $\pm$ 0.36	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	3s_vs_5z	2.37 $\pm$ 0.19	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3s_vs_5z	4.17 $\pm$ 0.15	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	3s_vs_5z	5.24 $\pm$ 0.59	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
MAPPO-Lag	3s_vs_5z	5.97 $\pm$ 0.65	3.02 $\pm$ 0.01	0.00 $\pm$ 0.00	1M	0
MAPPO-Lag	3s_vs_5z	7.82 $\pm$ 0.23	3.02 $\pm$ 0.02	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	3s_vs_5z	10.08 $\pm$ 0.56	3.01 $\pm$ 0.01	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s_vs_5z	16.88 $\pm$ 1.58	2.67 $\pm$ 0.40	0.15 $\pm$ 0.19	100k	0
SafePO	3s_vs_5z	3.52 $\pm$ 0.02	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
SafePO	3s_vs_5z	3.50 $\pm$ 0.01	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
SafePO	3s_vs_5z	4.32 $\pm$ 0.07	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	1
SafePO	3s_vs_5z	5.30 $\pm$ 0.40	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	1
SafePO	3s_vs_5z	6.41 $\pm$ 0.18	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	1
MAPPO-Lag	3s_vs_5z	4.20 $\pm$ 0.15	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
MAPPO-Lag	3s_vs_5z	5.51 $\pm$ 0.36	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	1
MAPPO-Lag	3s_vs_5z	6.61 $\pm$ 0.14	3.02 $\pm$ 0.02	0.00 $\pm$ 0.00	1M	1
MAPPO-Lag	3s_vs_5z	7.83 $\pm$ 0.10	3.02 $\pm$ 0.02	0.00 $\pm$ 0.00	2M	1
MAPPO-Lag	3s_vs_5z	9.33 $\pm$ 0.18	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
Safe Dreamers	3s_vs_5z	17.44 $\pm$ 1.49	2.67 $\pm$ 0.43	0.17 $\pm$ 0.22	100k	1
SafePO	8m	1.43 $\pm$ 0.01	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	8m	1.43 $\pm$ 0.02	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	8m	0.95 $\pm$ 0.06	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	8m	0.52 $\pm$ 0.03	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	8m	0.33 $\pm$ 0.05	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	8m	2.90 $\pm$ 0.02	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	8m	9.09 $\pm$ 0.03	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
MAPPO-Lag	8m	16.56 $\pm$ 0.83	5.67 $\pm$ 0.46	0.60 $\pm$ 0.08	1M	0
MAPPO-Lag	8m	19.84 $\pm$ 0.06	2.42 $\pm$ 0.18	0.98 $\pm$ 0.00	2M	0
MAPPO-Lag	8m	19.96 $\pm$ 0.02	0.90 $\pm$ 0.17	1.00 $\pm$ 0.00	5M	0
Safe Dreamers	8m	18.64 $\pm$ 2.02	2.81 $\pm$ 1.91	0.86 $\pm$ 0.21	100k	0
SafePO	8m	1.63 $\pm$ 0.01	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4

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Table 1 – continued

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
SafePO	8m	1.63 $\pm$ 0.02	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
SafePO	8m	2.66 $\pm$ 0.19	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
SafePO	8m	2.94 $\pm$ 0.50	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	4
SafePO	8m	1.95 $\pm$ 0.40	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	4
MAPPO-Lag	8m	2.77 $\pm$ 0.15	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
MAPPO-Lag	8m	9.04 $\pm$ 0.43	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
MAPPO-Lag	8m	15.86 $\pm$ 2.19	5.96 $\pm$ 1.20	0.53 $\pm$ 0.25	1M	4
MAPPO-Lag	8m	19.75 $\pm$ 0.20	2.70 $\pm$ 0.54	0.97 $\pm$ 0.03	2M	4
MAPPO-Lag	8m	19.95 $\pm$ 0.03	1.10 $\pm$ 0.36	0.99 $\pm$ 0.00	5M	4
Safe Dreamers	8m	17.26 $\pm$ 1.04	5.03 $\pm$ 0.42	0.69 $\pm$ 0.10	100k	4
SafePO	MMM	1.87 $\pm$ 0.01	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
SafePO	MMM	1.86 $\pm$ 0.03	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
SafePO	MMM	1.49 $\pm$ 0.07	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
SafePO	MMM	1.14 $\pm$ 0.12	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	0
SafePO	MMM	0.79 $\pm$ 0.08	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	0
MAPPO-Lag	MMM	2.63 $\pm$ 0.14	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
MAPPO-Lag	MMM	6.44 $\pm$ 0.49	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
MAPPO-Lag	MMM	10.58 $\pm$ 0.23	10.11 $\pm$ 0.01	0.00 $\pm$ 0.00	1M	0
MAPPO-Lag	MMM	12.58 $\pm$ 0.31	10.06 $\pm$ 0.02	0.01 $\pm$ 0.00	2M	0
MAPPO-Lag	MMM	19.89 $\pm$ 0.14	0.97 $\pm$ 0.61	0.99 $\pm$ 0.02	5M	0
Safe Dreamers	MMM	17.05 $\pm$ 2.36	6.77 $\pm$ 1.79	0.59 $\pm$ 0.25	100k	0
SafePO	MMM	1.89 $\pm$ 0.04	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4
SafePO	MMM	1.88 $\pm$ 0.04	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
SafePO	MMM	1.47 $\pm$ 0.03	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
SafePO	MMM	1.14 $\pm$ 0.04	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1M	4
SafePO	MMM	0.73 $\pm$ 0.01	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	2M	4
MAPPO-Lag	MMM	2.61 $\pm$ 0.22	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
MAPPO-Lag	MMM	6.29 $\pm$ 1.28	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
MAPPO-Lag	MMM	10.04 $\pm$ 0.92	10.11 $\pm$ 0.02	0.00 $\pm$ 0.00	1M	4
MAPPO-Lag	MMM	12.66 $\pm$ 0.84	9.92 $\pm$ 0.32	0.03 $\pm$ 0.06	2M	4
MAPPO-Lag	MMM	19.61 $\pm$ 0.47	1.41 $\pm$ 1.07	0.97 $\pm$ 0.03	5M	4
Safe Dreamers	MMM	17.02 $\pm$ 1.32	7.88 $\pm$ 0.16	0.55 $\pm$ 0.07	100k	4
SafePO	bane_vs_bane	19.26 $\pm$ 0.01	22.55 $\pm$ 0.02	0.79 $\pm$ 0.01	5M	0
SafePO	bane_vs_bane	19.27 $\pm$ 0.03	22.45 $\pm$ 0.05	0.76 $\pm$ 0.02	100k	0
SafePO	bane_vs_bane	19.70 $\pm$ 0.05	18.54 $\pm$ 0.20	0.91 $\pm$ 0.01	500k	0
SafePO	bane_vs_bane	19.80 $\pm$ 0.04	14.08 $\pm$ 0.16	0.93 $\pm$ 0.01	1M	0
SafePO	bane_vs_bane	19.86 $\pm$ 0.02	9.93 $\pm$ 0.18	0.95 $\pm$ 0.00	2M	0
MAPPO-Lag	bane_vs_bane	—	—	—	100k	0
MAPPO-Lag	bane_vs_bane	—	—	—	500k	0
MAPPO-Lag	bane_vs_bane	—	—	—	1M	0
MAPPO-Lag	bane_vs_bane	—	—	—	2M	0
MAPPO-Lag	bane_vs_bane	—	—	—	5M	0
Safe Dreamers	bane_vs_bane	17.75 $\pm$ 2.82	13.91 $\pm$ 7.11	0.62 $\pm$ 0.39	100k	0
SafePO	bane_vs_bane	19.37 $\pm$ 0.09	22.40 $\pm$ 0.06	0.80 $\pm$ 0.02	5M	4
SafePO	bane_vs_bane	19.38 $\pm$ 0.07	22.41 $\pm$ 0.06	0.80 $\pm$ 0.01	100k	4
SafePO	bane_vs_bane	19.71 $\pm$ 0.03	18.69 $\pm$ 0.28	0.92 $\pm$ 0.01	500k	4
SafePO	bane_vs_bane	19.74 $\pm$ 0.02	13.67 $\pm$ 0.56	0.92 $\pm$ 0.01	1M	4
SafePO	bane_vs_bane	19.92 $\pm$ nan	9.49 $\pm$ nan	0.97 $\pm$ nan	2M	4
MAPPO-Lag	bane_vs_bane	—	—	—	100k	4
MAPPO-Lag	bane_vs_bane	—	—	—	500k	4
MAPPO-Lag	bane_vs_bane	—	—	—	1M	4
MAPPO-Lag	bane_vs_bane	—	—	—	2M	4
MAPPO-Lag	bane_vs_bane	—	—	—	5M	4

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Table 1 – continued

	Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
Safe Dreamers		bane_vs_bane	19.91 ± 0.11	6.37 ± 1.20	0.98 ± 0.02	100k	4